In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```
df = pd.read_csv('E_Commerce.csv')
```

In [4]:
```
df
```

Out[4]:

	order_id	order_date	state	zone	category	brand_type	customer_
0	ORD0000001	2023-01-31	West Bengal	East	Fashion	Mass	
1	ORD0000002	2023-01-31	Gujarat	West	Sports & Fitness	Mass	
2	ORD0000003	2023-01-31	Delhi NCR	North	Grocery Essentials	Mass	
3	ORD0000004	2023-01-31	Madhya Pradesh	Central	Footwear	Mass	
4	ORD0000005	2023-01-31	Haryana	North	Fashion	Premium	
...	...	...	...	...	...	...	
30595	ORD0030596	2025-12-31	Uttar Pradesh	North	Premium Lifestyle	Mass	
30596	ORD0030597	2025-12-31	Haryana	North	Footwear	Premium	
30597	ORD0030598	2025-12-31	Odisha	East	Footwear	Mass	
30598	ORD0030599	2025-12-31	West Bengal	East	Sports & Fitness	Mass	
30599	ORD0030600	2025-12-31	Maharashtra	West	Footwear	Premium	

30600 rows × 16 columns

In [5]:
```
df.shape
```

Out[5]: (30600, 16)

In [7]:
```
df.head()
```

Out[7]:

	order_id	order_date	state	zone	category	brand_type	customer_gender
0	ORD0000001	2023-01-31	West Bengal	East	Fashion	Mass	Male
1	ORD0000002	2023-01-31	Gujarat	West	Sports & Fitness	Mass	Male
2	ORD0000003	2023-01-31	Delhi NCR	North	Grocery Essentials	Mass	Male
3	ORD0000004	2023-01-31	Madhya Pradesh	Central	Footwear	Mass	Female
4	ORD0000005	2023-01-31	Haryana	North	Fashion	Premium	Female

In [8]:

```
df.tail()
```

Out[8]:

	order_id	order_date	state	zone	category	brand_type	customer_g
30595	ORD0030596	2025-12-31	Uttar Pradesh	North	Premium Lifestyle	Mass	
30596	ORD0030597	2025-12-31	Haryana	North	Footwear	Premium	
30597	ORD0030598	2025-12-31	Odisha	East	Footwear	Mass	F
30598	ORD0030599	2025-12-31	West Bengal	East	Sports & Fitness	Mass	F
30599	ORD0030600	2025-12-31	Maharashtra	West	Footwear	Premium	

In [9]:

```
df.head(10)
```

Out[9]:

	order_id	order_date	state	zone	category	brand_type	customer_gender
0	ORD0000001	2023-01-31	West Bengal	East	Fashion	Mass	Male
1	ORD0000002	2023-01-31	Gujarat	West	Sports & Fitness	Mass	Male
2	ORD0000003	2023-01-31	Delhi NCR	North	Grocery Essentials	Mass	Male
3	ORD0000004	2023-01-31	Madhya Pradesh	Central	Footwear	Mass	Female
4	ORD0000005	2023-01-31	Haryana	North	Fashion	Premium	Female
5	ORD0000006	2023-01-31	Punjab	North	Grocery Essentials	Mass	Female
6	ORD0000007	2023-01-31	Delhi NCR	North	Home & Living	Mass	Female
7	ORD0000008	2023-01-31	West Bengal	East	Fashion	Mass	Male
8	ORD0000009	2023-01-31	West Bengal	East	Grocery Essentials	Mass	Male
9	ORD0000010	2023-01-31	Gujarat	West	Grocery Essentials	Premium	Male

In [10]:
```
df.tail(7)
```

Out[10]:

	order_id	order_date	state	zone	category	brand_type	customer_(
30593	ORD0030594	2025-12-31	Karnataka	South	Premium Lifestyle	Mass	
30594	ORD0030595	2025-12-31	Madhya Pradesh	Central	Home & Living	Mass	
30595	ORD0030596	2025-12-31	Uttar Pradesh	North	Premium Lifestyle	Mass	
30596	ORD0030597	2025-12-31	Haryana	North	Footwear	Premium	
30597	ORD0030598	2025-12-31	Odisha	East	Footwear	Mass	
30598	ORD0030599	2025-12-31	West Bengal	East	Sports & Fitness	Mass	
30599	ORD0030600	2025-12-31	Maharashtra	West	Footwear	Premium	

In [11]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 30600 entries, 0 to 30599
Data columns (total 16 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   order_id              30600 non-null  str
 1   order_date            30600 non-null  str
 2   state                 30600 non-null  str
 3   zone                  30600 non-null  str
 4   category              30600 non-null  str
 5   brand_type            30600 non-null  str
 6   customer_gender       30600 non-null  str
 7   customer_age          30600 non-null  int64
 8   base_price            30600 non-null  float64
 9   discount_percent      30600 non-null  float64
 10  final_price           30600 non-null  float64
 11  units_sold            30600 non-null  int64
 12  revenue               30600 non-null  float64
 13  sales_event           30600 non-null  str
 14  competition_intensity 30600 non-null  str
 15  inventory_pressure    30600 non-null  str
dtypes: float64(4), int64(2), str(10)
memory usage: 3.7 MB
```

In [12]:
```
df.columns
```

Out[12]:
```
Index(['order_id', 'order_date', 'state', 'zone', 'category', 'brand_type',
       'customer_gender', 'customer_age', 'base_price', 'discount_percent',
       'final_price', 'units_sold', 'revenue', 'sales_event',
       'competition_intensity', 'inventory_pressure'],
      dtype='str')
```

In [13]:
```
df[['state']]
```

Out[13]:

	state
0	West Bengal
1	Gujarat
2	Delhi NCR
3	Madhya Pradesh
4	Haryana
...	...
30595	Uttar Pradesh
30596	Haryana
30597	Odisha
30598	West Bengal
30599	Maharashtra

30600 rows × 1 columns

In [14]:
```
df[['state','zone']]
```

Out[14]:

	state	zone
0	West Bengal	East
1	Gujarat	West
2	Delhi NCR	North
3	Madhya Pradesh	Central
4	Haryana	North
...	...	...
30595	Uttar Pradesh	North
30596	Haryana	North
30597	Odisha	East
30598	West Bengal	East
30599	Maharashtra	West

30600 rows × 2 columns

In [15]:
```
print(df.isnull().sum())
```

```
order_id                0
order_date              0
state                   0
zone                    0
category                0
brand_type              0
customer_gender         0
customer_age            0
base_price              0
discount_percent        0
final_price             0
units_sold              0
revenue                 0
sales_event             0
competition_intensity   0
inventory_pressure      0
dtype: int64
```

In [16]:
```
df.duplicated().sum()
```

Out[16]:
```
np.int64(0)
```

In [17]:
```
df[['state']][4:15]
```

Out[17]:

	state
4	Haryana
5	Punjab
6	Delhi NCR
7	West Bengal
8	West Bengal
9	Gujarat
10	Uttar Pradesh
11	Kerala
12	Karnataka
13	Maharashtra
14	Odisha

In [18]:
```
df[['inventory_pressure']][4:16:2]
```

Out[18]:

	inventory_pressure
4	High
6	High
8	Low
10	Low
12	High
14	Low

In [19]:
```
df.dtypes
```

Out[19]:
```
order_id                  str
order_date                str
state                     str
zone                      str
category                  str
brand_type                str
customer_gender           str
customer_age            int64
base_price            float64
discount_percent      float64
final_price           float64
units_sold              int64
revenue               float64
sales_event               str
competition_intensity     str
inventory_pressure        str
dtype: object
```

In [20]:
```
df.describe()
```

Out[20]:

	customer_age	base_price	discount_percent	final_price	units_sold	
count	30600.000000	30600.000000	30600.000000	30600.000000	30600.000000	306
mean	35.568268	3898.637966	38.111218	2415.015197	31.705490	710
std	12.872780	3371.290126	16.608485	2264.892146	21.381121	836
min	18.000000	100.030000	0.010000	35.010000	2.000000	2
25%	25.000000	1314.637500	24.840000	747.010000	16.000000	166
50%	33.000000	2975.015000	37.290000	1728.605000	27.000000	421
75%	45.000000	5342.480000	51.840000	3309.630000	42.000000	925
max	65.000000	14999.170000	65.000000	14682.280000	161.000000	9061

In [ ]: